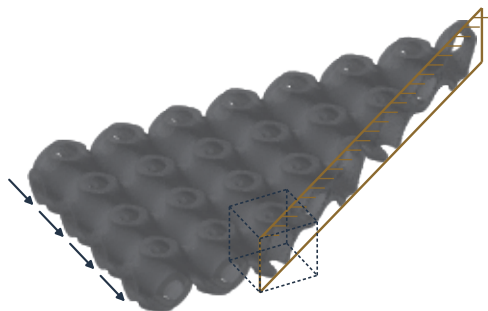
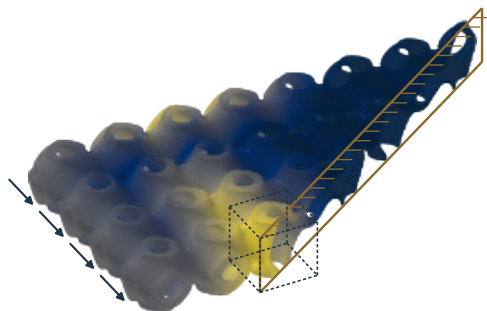


(a) Uniform start, $\tau = 0.40$



NICE $C = 75.59$, exact $C = 75.60$

(b) Final NICE design (iteration 23)

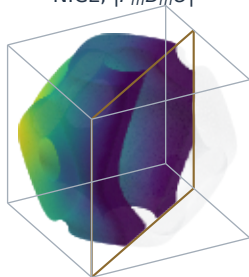


NICE $C = 78.26$, exact $C = 78.29$

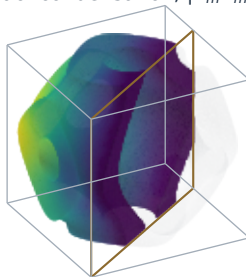
0.18 0.30 0.40 0.50 0.60 0.69
local thickness parameter $\tau(x)$ on the walls

(c) Cut cell (3, 2, 0) at the final design, design load: displacement magnitude on the walls

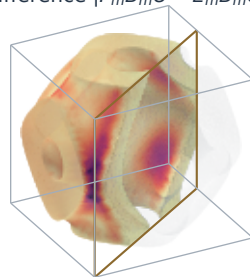
NICE, $|F_m B_m \hat{U}|$



exact condensation, $|E_m B_m U|$



difference $|F_m B_m \hat{U} - E_m B_m U|$



0 2 4 6 8 10 12
displacement magnitude $|u|$ (shared scale)

0.00 0.01 0.02 0.03
 $|u_{\text{NICE}} - u_{\text{exact}}|$ (own scale)

Relative difference of the cell field, NICE vs exact: 1.04% in the energy norm, 0.76% in the nodal ℓ_2 norm;
largest difference on the walls 0.29% of the largest $|u|$ on the walls